AI BUILDER CUP / BFSI

ShieldIQ X

The Autonomous Financial Crime Investigation Platform

From a suspicious transaction to an evidence-led, human-reviewed investigation.

Team Byte Us | Google Cloud AI Builder Cup
Synthetic-data MVP | Evidence-first | Human accountable

No production deployment, live model, or measured ROI is claimed.

An alert is not an investigation

Financial crime teams need to turn fragmented signals into a reviewable decision.

Fragmented signals

Transactions, accounts, devices, network observations, and prior alerts can live in separate views.

Manual reconstruction

Investigators spend effort rebuilding chronology, relationships, and supporting context.

Regulatory burden

Reports need traceable facts, source references, unknowns, and accountable review.

Delayed decisions

Handoffs and repeated evidence gathering can slow triage; local pilot evidence is needed to quantify it.

Measure impact - do not guess it

The opportunity is clear; dollar savings and market-size claims require institution data.

Financial exposure

Potential impact area

Measure: alert-to-triage time and case aging.

Investigation cost

Potential impact area

Measure: analyst minutes per case and evidence rework.

Operational friction

Potential impact area

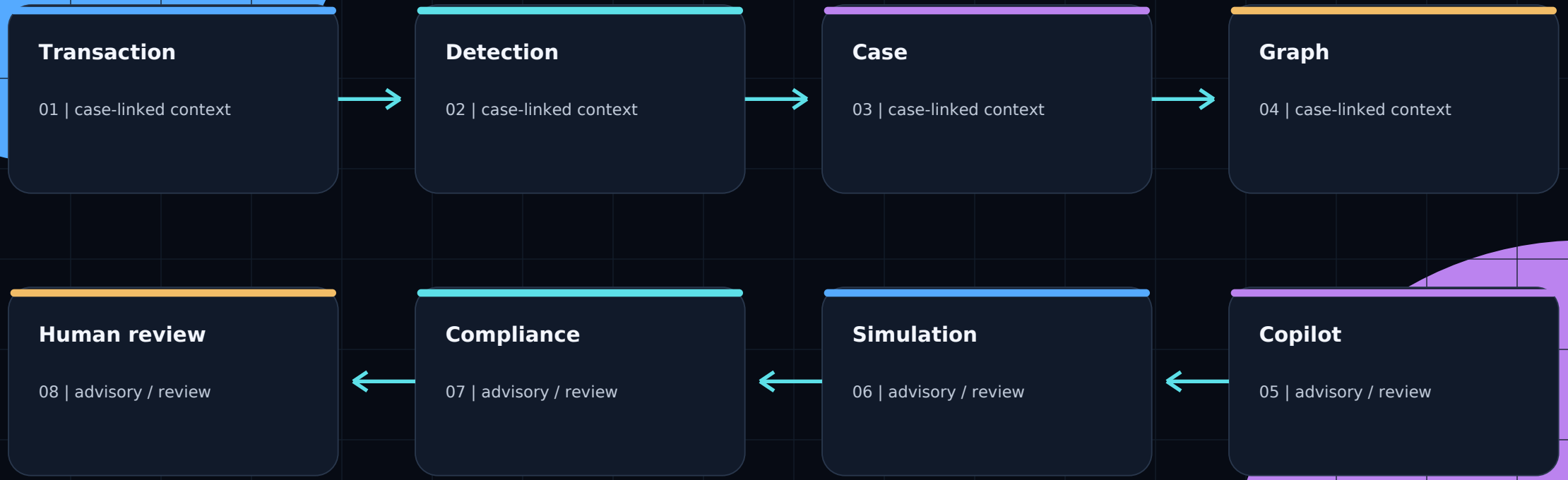
Measure: handoffs, report corrections, citation coverage.

PILOT BASELINE REQUIRED

No independently verified fraud-loss reduction, ROI, or productivity figure was available in the repository.

One governed investigation loop

Existing workflows stay intact; evidence and human review connect the journey.

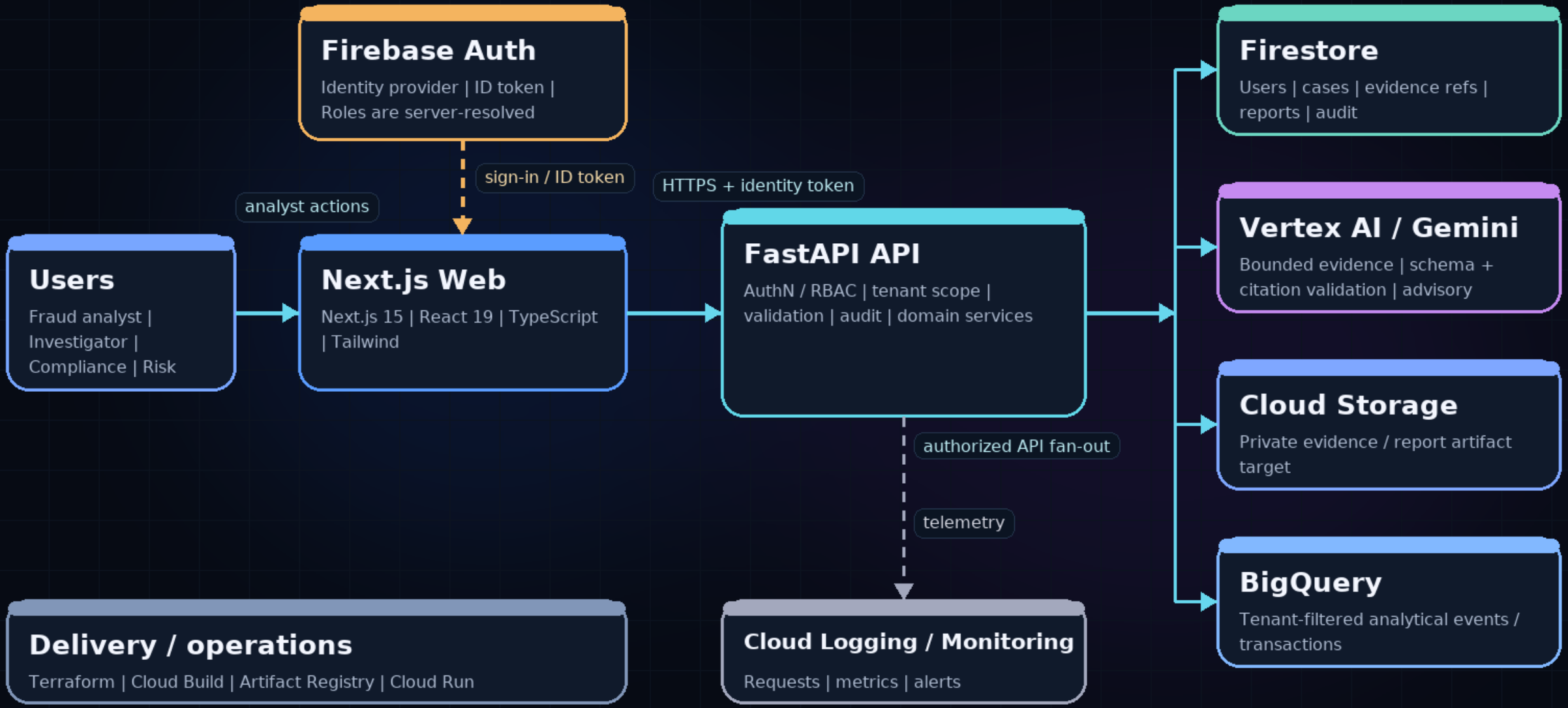


AI ADVISORY - HUMAN DECISION

SHIELDIQ X / SYSTEM ARCHITECTURE

TARGET ONLY - NOT CLOUD-VERIFIED

Evidence-first financial crime investigation | logical repository view



API is the authorization boundary. Firestore client rules deny direct access. Human approval precedes report export.

Repository source/configuration diagram only - cloud resources and third-party integrations have not been deployed or verified.

Explainable signals, not a verdict

Versioned rule logic contributes only when a signal is linked to authorized evidence.

TESTED SYNTHETIC CASE

87

TRIAGE SCORE / RULES V1

12 base + 30 velocity + 25 shared device + 20 new beneficiary

Rapid velocity

Transaction evidence supports a burst of pass-through activity.

Shared device

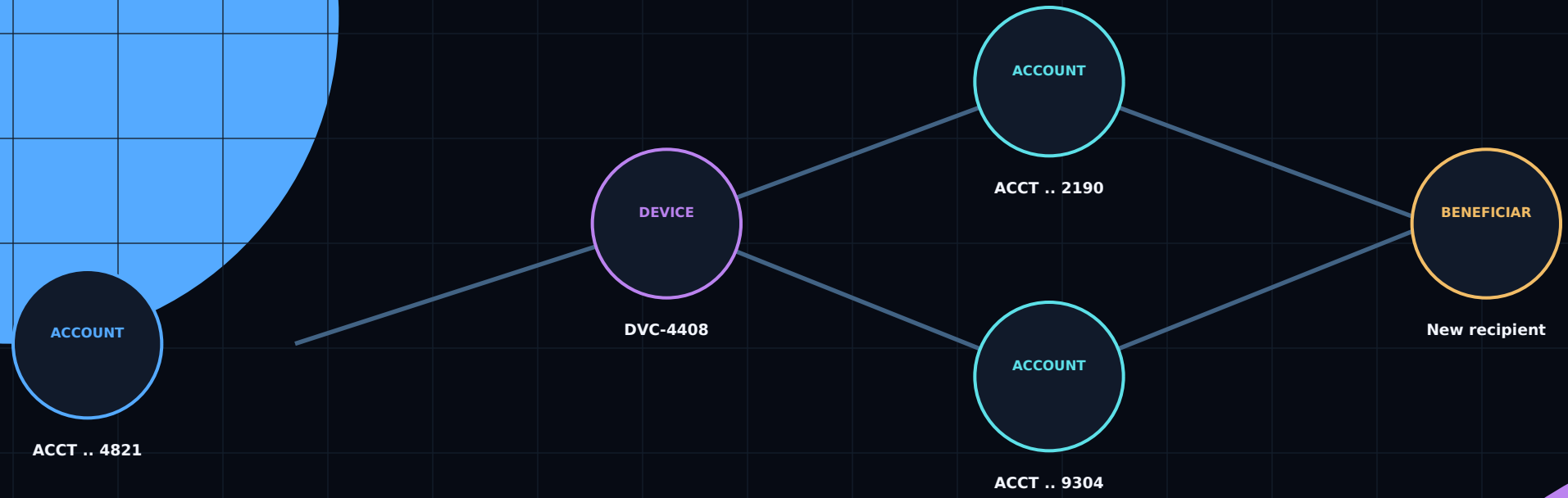
The evidence-linked device relationship adds context, not proof of common ownership.

New beneficiary

First-seen counterparty adds a review reason. Unmapped signals are not silently scored.

Follow the evidence through the network

A shared attribute is an investigative lead - not proof of common control.



Provenance matters

Every relationship should resolve to a source reference, observed time, and context. The API graph-filter test passes; full browser, accessibility, and staging verification remain open.

Structured assistance with visible boundaries

A generated draft is useful only when the investigator can inspect its evidence and uncertainty.

RESPONSE SHAPE - ILLUSTRATIVE, NOT LIVE OUTPUT

Observed facts
Claims with authorized evidence IDs

Unknowns
Missing or conflicting context

Recommendations
Suggested checks for human review

Provider status

Live Vertex AI generation was not verified in this review. Show the explicit unavailable state unless a real staging request succeeds.

Guardrails

Authorized evidence only. Mask sensitive text. Validate output schema and citations. Human review remains required.

Explore a scenario - not a forecast

The numeric result is a bounded heuristic with assumptions, not a calibrated loss prediction.

Inputs

Horizon: 1-365 days
Exposure: bounded amount
Containment: 0 to 1

Review output

Scenario estimate
Assumptions and uncertainty
Optional AI interpretation is separate

Guardrail

Keep scenario parameters, assumptions, and caveats on screen. Do not describe the value as certain future loss or an automated intervention.

Draft, review, approve, then export

The product prepares reviewable records; it does not file a SAR automatically.



Verified policy checks

Automated tests reject export of an unapproved draft and reject self-approval by the report creator. Live generation, approval, and object-store export still need staging evidence.

Managed-service design; scale still to prove

The source architecture describes intended boundaries, not measured throughput or availability.

Firebase Auth

Identity

Cloud integration NOT VERIFIED

Cloud Run

Web + API

Cloud integration NOT VERIFIED

Firestore

Operational cases

Cloud integration NOT VERIFIED

BigQuery

Analytics

Cloud integration NOT VERIFIED

Cloud Storage

Artifacts

Cloud integration NOT VERIFIED

Vertex AI

Gemini

Cloud integration NOT VERIFIED

Before claiming scale: benchmark latency, quotas, costs, concurrency, recovery objectives, and tenant isolation in an approved staging environment.

Earn trust before claiming impact

A measured, governed pilot is the path from MVP to institution-ready capability.

1 Prove

Staging identity, tenant isolation, citations, approvals, deployment and rollback.

2 Measure

Triage time, evidence completeness, report rework, citation coverage, case aging.

3 Govern

Institution-specific model risk, retention, privacy, audit integrity, regulatory review.

From a signal to a reviewable investigation.

The human remains accountable.

Next milestone: a synthetic-data staging pilot with verified integrations and measured outcomes. No loss-reduction or ROI figure is claimed.